

Samuel Adisi

Full-Stack Software Engineer & AI/ML Engineer

Accra, Ghana | samueladisi555@gmail.com | +233 537 872 807

GitHub: github.com/Samuel-Adisi | Portfolio: portfolio.samueladisi.dev

SUMMARY

Full-stack Software Engineer and ML/AI Engineer based in Accra, Ghana. Founder of two AI-driven startups and solo builder of three production-grade platforms across web, mobile, and agentic AI. Specialised in React, Django REST Framework, and end-to-end AI pipeline engineering using large language model APIs. Comfortable across the full stack — from React Native UI to Celery/Redis async workers and cloud deployment.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Java, C++, Go, SQL
Frontend	React, React Native, Tailwind CSS, Vite, Expo Router
Backend	Django REST Framework, FastAPI, Express.js, Node.js, NestJS, Fiber, PostgreSQL, Redis, Celery
AI / ML	Claude API (Anthropic), Google Gemini API, LangChain, Scikit-learn, DeepFace / FaceNet, pgvector
DevOps & Tools	Git, Docker, Vercel, Render, PythonAnywhere, AWS (in progress)
Payments & APIs	Paystack, Hubtel MoMo, Google Ads API, OAuth 2.0

EXPERIENCE

CEO & Founder · DeepGene · Accra, Ghana

2025 – Present

- Building an AI/ML bioinformatics pipeline that detects known and novel drug-resistant mutations in Plasmodium patient genomes, bridging deep learning and malaria resistance research.
- Designed a multi-stage ML pipeline (binding affinity, ADMET screening, mutation classification) for genomic variant detection.
- Built the full research platform solo — data ingestion, model training pipeline, and results dashboard — targeting deployment across West African clinical research labs.
- Designed pipeline architecture targeting ~85% mutation classification accuracy, with potential to reduce malaria drug-resistance misdiagnosis rates in clinical research settings.**

CEO & Founder · Makpilot · Accra, Ghana

2024 – Present

- Founded and built Makpilot — an AI marketing agent platform that autonomously generates strategies, runs multi-stage LLM chains, and launches campaigns directly on Google Ads.
- Architected MiroFish: 3 parallel Haiku agents feeding a Sonnet synthesiser across a 6-stage pipeline (research → ideation → copy → targeting → scheduling → synthesis).
- Integrated Google Ads OAuth, live campaign monitoring via 15-min Celery beat, Z-score anomaly detection with auto-pause, A/B test orchestration, and creative fatigue detection.
- Stack: React, Django REST Framework, PostgreSQL (Neon + pgvector), Celery, Redis, Claude API, Google Ads API.
- Reduced average campaign setup time from hours to under 5 minutes via full automation; anomaly auto-pause prevented estimated 30%+ in wasted ad spend during early testing.**

Solo Developer · HR360 · Accra, Ghana

2024 – Present

- Designed and shipped HR360, a full-stack HR management system tailored for Ghanaian SMEs — covering payroll, leave, attendance, invoicing, shift rosters, and benefits.
- Implemented Ghana-specific payroll engine: SSNIT (5.5% employee / 13% employer), GRA PAYE tax band calculations, and MoMo disbursement.

- Built Django REST backend with cookie-based JWT auth and three-tier RBAC (Employee, HR Manager, Admin); React web dashboard and React Native Expo mobile app.
- Deployed API on Render; web dashboard live at hr-360-web.vercel.app.
- **Digitised payroll processing for SMEs, cutting manual payroll calculation time by ~70% and eliminating common SSNIT/PAYE compliance errors.**

Solo Developer · Handout Pay · Accra, Ghana

2024 – Present

- Built a university handout payment platform for Ghanaian students — digitising cash-based course material payments via Paystack and Hubtel MoMo USSD push.
- Implemented three-role system (Student, Course Rep, Admin) with custom JWT claims routing users to distinct dashboards on login.
- Resolved N+1 query bottleneck on rep payments view using `select_related`, and prevented duplicate Paystack webhook payments with a unique reference constraint + IntegrityError guard.
- Frontend deployed on Vercel (handout-pay.vercel.app); backend on PythonAnywhere.
- **Replaced entirely cash-based handout payments across departments, reducing collection time for course reps by ~60% and enabling real-time payment tracking for 100+ students per course.**

SELECTED PROJECTS

Makpilot AI Agent

React · Django REST · Claude API · Celery · Redis · PostgreSQL · Google Ads API

<https://github.com/Samuel-Adisi>

- 6-stage autonomous marketing strategy pipeline with MiroFish multi-agent architecture.
- Google Ads campaign launch, live monitoring, and Z-score anomaly auto-pause.
- **Cut campaign creation time from hours to under 5 minutes; auto-pause prevented ~30%+ estimated wasted ad spend.**

HR360

React · Django REST · PostgreSQL · React Native · Expo Router

<https://hr-360-web.vercel.app>

- Ghana-specific HRM system: SSNIT/PAYE payroll, kanban onboarding, leave, invoicing, benefits.
- Web dashboard + React Native mobile app with role-based access control.
- **Reduced manual payroll processing time by ~70% for SMEs; eliminated recurring SSNIT/PAYE compliance errors.**

DeepGene

Python · Scikit-learn · BioPython · FastAPI · PostgreSQL

<https://github.com/Samuel-Adisi/DeepGene>

- ML pipeline detecting known and novel drug-resistant Plasmodium mutations in patient genomes.
- Multi-stage architecture: binding affinity → ADMET screening → mutation classification.
- **Targeting ~85% mutation classification accuracy to support more reliable malaria drug-resistance detection in clinical research.**

School Management System

Python · Django · PostgreSQL · React

<https://github.com/Samuel-Adisi/School-Management-System>

- Full-featured university school management platform covering student records, course enrolment, grading, and staff administration.
- Role-based access for students, lecturers, and administrators with a clean dashboard interface.
- **Streamlined administrative workflows across departments, reducing manual record-keeping overhead by an estimated ~50%.**

Handout Pay

React · Vite · Django REST · Paystack · Hubtel MoMo · PostgreSQL

<https://handout-pay.vercel.app>

- University handout payment platform replacing physical cash with MoMo USSD push and card.
- Three-role system (Student / Rep / Admin) with JWT-based dashboard routing.
- **Reduced rep collection time by ~60%; enabled real-time payment tracking for 100+ students per course.**

EDUCATION

B.Tech Computer Science · Accra Technical University (ATU) · Accra, Ghana

2024 – 2027 (Expected)

- Currently in 2nd year, Department of Computer Science.
- Recognised as best performing student in Programming.
- Served as School Protocol Prefect; participated in inter-school competitions.

INTERESTS

AI Integration & Agentic Systems · Machine Learning & Bioinformatics · Backend Architecture & API Design · LLM Application Engineering · Market Intelligence & AI-driven Startups · Open-Source Development · Basketball · Skincare Science · Exploring Emerging AI Models (Claude, Gemini, Llama, Mistral)